

SMART INDIA HACKATHON 2026

NODAL CONNECTOR (WORKING MVP)

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❑ **Problem Statement ID : SIH26044**

❑ **Problem Statement Title :** Portal for Academia - Industry collaboration for Skill Mapping, Internships and Placement

❑ **Theme : Smart Automation**

❑ **PS Category : Software**

❑ **Project Status : Functional MVP Prototype (Full-Stack Built & Deployed)**

❑ **Team ID :** [Insert Team ID]

❑ **Team Name (Registered on portal) :** [Insert Registered Team Name]

WORKING MVP HIGHLIGHTS

- 🌐 Live Web Portal:
Deployed frontend & backend API
- 🎯 4 Functional Roles:
Student, Industry, Faculty, Admin
- 🧠 AI Diagnostic Tests:
Dynamic skill testing (Groq LLaMA 3.3)
- 📺 Live WebRTC Rooms:
Zero-install video & collaborative whiteboard
- 📊 Dynamic Skill Match:
Transparent score based on real abilities

NODAL CONNECTOR

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❖ Proposed Solution

Pillar 01: Smart Skill Diagnostic & Gap Remediation

- Objective AI Diagnostic Assessment (Groq LLaMA 3.3) assessing real student competencies with authentic 0-100% scoring.
- UGC Section 22 Degree Normalizer with 500ms debounce matching engineering, bio-pharma, and healthcare degrees.
- Automated Skill Gap Matrix linking identified weaknesses directly to curated roadmap.sh learning pathways.
- Real Cohort Percentile Benchmarks calculated dynamically from peer test attempts without artificial score boosts.

Pillar 02: Unified Opportunities & Direct In-App Hiring

- Centralized Marketplace for verified experiential internships and entry-level jobs with stipends.
- Transparent Compatibility Score based on verified skills matching specific posting requirements.
- Real-Time Application Lifecycle: Applied → Under Review → Interview Scheduled → Offered / Rejected.
- Collaborative Industry Learning Programs: Companies sponsor structured modules for talent upskilling.

[REAL SCREENSHOT: STUDENT DASHBOARD & AI ASSESSMENT]

Route URL: </student/dashboard> & </student/assessment>

Take a screenshot of the Student Dashboard showing the Skill Radar Chart, Active Applications, or the Diagnostic Test in action!

TECHNICAL APPROACH

❖ Technology Stack

Frontend (React 18 + Vite + Tailwind CSS)

- Single Page App (SPA) with Protected Routes for 4 personas.
- Zustand state store for user sessions & real-time alerts.
- Dynamic SVG skill radar charts & debounced UGC degree selector.

Backend & Real-Time (Node.js + Express + Socket.IO)

- Modular RESTful API with JWT authentication & role authorization.
- Socket.IO signaling for browser-to-browser WebRTC video/audio calls.
- Real-time synchronized whiteboard canvas for technical discussions.

AI & Database (Groq LLaMA 3.3 + MongoDB Atlas)

- Groq LPU high-speed inference for instant diagnostic question generation.
- 17 Mongoose schemas with compound indexes for fast applicant search.
- roadmap.sh curriculum integration for automated student study roadmaps.

[REAL SCREENSHOT: LIVE WEBRTC INTERVIEW & WHITEBOARD]

Route URL: /meetings (or direct interview room)

Take a screenshot of the Live Meeting page showing the WebRTC video tiles, controls, and the interactive drawing canvas board!

FEASIBILITY AND VIABILITY

Feasibility

- Technical: Open web standards (React, Node, WebRTC) needing 0 external downloads.
- Operational: Lightweight UI runs on mobile devices and basic college Wi-Fi.
- Financial: Serverless cloud hosting keeps running costs minimal for the MVP.

Viability

- High College Adoption: Helps placement cells connect students to real recruiters.
- Recruiter Incentive: Pre-assessed candidates with transparent skill matches.
- MVP Scalability: Clean modular code easily transitions into full production.

Challenges

- Cold Start: Students having unverified or incomplete skill profiles.
- Curricular Lag: University syllabi updating slower than industry tech.
- Campus Firewalls: Restrictive college NAT networks blocking P2P video.

Use Cases

- Student Career Discovery: Student identifies a skill gap in bio-pharma / coding, follows the roadmap, and applies for an internship.
- Recruiter Sourcing: Employer filters candidates by degree & verified skill compatibility, scheduling an instant video interview.
- Faculty Collaboration: Professor browses industry research immersion programs and schedules student workshops.

Solutions

- AI Diagnostic Testing: Fast 10-question evaluation immediately verifies core skills.
- Industry Learning Tracks: Companies post custom learning modules to bridge syllabus lags.
- STUN/TURN Fallback: Integrated WebRTC fallback ensures video calls connect even on campus Wi-Fi.

[REAL SCREENSHOT: INTERNSHIP & JOB MARKETPLACE]

Route URL: `/student/internships` or `/industry/post`

Take a screenshot of the Internships portal showing opportunity cards, stipend badges, and the 'Skill Match' indicator!

IMPACT AND BENEFITS

❖ Benefits Of The Solution

For Students & Jobseekers

- Authentic skill feedback with personalized learning roadmaps.
- Direct access to verified internship & job openings with stipends.
- Shareable digital portfolio of verified skills.

For Industry & Recruiters

- Saves hours of screening through transparent skill-match scores.
- Schedule and conduct technical interviews directly in the portal.
- Sponsor tailored learning programs to groom talent.

For Academic Faculty

- Real-time insight into in-demand industry competencies.
- Direct channel to connect students with corporate mentors.

❖ Potential Impact On The Target Audience

Bridging the Employability Gap

Helps students uncover what they don't know early in college, giving them actionable steps to become job-ready.

Democratizing Tier-2/3 College Access

Gives students from smaller colleges direct visibility to companies without relying solely on campus placement drives.

Multi-Disciplinary Synergy

Connects traditional sciences, Ayush bio-pharma, and computer engineering under one unified skill taxonomy.

Transparent Merit-Based Hiring

Replaces inflated resumes with verified diagnostic results and real project showcases.

[REAL SCREENSHOT: RECRUITER APPLICANT MANAGEMENT]

Route URL: </industry/applicants> or </industry/candidates>

Take a screenshot of the Recruiter page showing candidate applications, compatibility score pills, and interview actions!

RESEARCH AND REFERENCES

• Details / Links of the reference and research work

1. National Education Policy (NEP 2020) — Ministry of Education

Reference: NEP 2020 Chapter 11 on vocational training, industry linkage, and experiential learning credits.

https://www.education.gov.in/sites/upload_files/mhrd/files/NEP_Final_English_0.pdf

2. University Grants Commission (UGC) Degree Specification

Reference: UGC Act 1956 Section 22 official nomenclature standardization framework, utilized in our degree selector.

https://www.ugc.gov.in/pdfnews/spec_degrees.pdf

3. AICTE National Internship Policy & Guidelines

Reference: Framework for mandatory technical internships and industry-academia credit integration.

<https://internship.aicte-india.org/>

4. roadmap.sh Open-Source Curriculum Framework

Reference: Community developer skill roadmaps used to curate learning paths for student remediation.

<https://github.com/kamranahmedse/developer-roadmap>

5. Working MVP Repository & Live Deployments (SIH26044)

- Live Client: <https://sih26044-ayush-portal.vercel.app>
- Live Backend API: <https://sih26044-ayush-portal-production.up.railway.app>
- GitHub: <https://github.com/adisharma9548/sih26044-ayush-portal>

Working prototype solving SIH26044 with full MongoDB Atlas persistence and live WebRTC signaling.

[REAL SCREENSHOT: INSTITUTIONAL PROGRESS & ANALYTICS]

Route URL: </admin/reports> or
</admin/progress>

Take a screenshot of the Institutional Admin analytics dashboard showing student placement trends and skill distributions!